

Frugal Innovation in Emerging Markets

Marco Zeschky, Bastian Widenmayer & Oliver Gassmann

To cite this article: Marco Zeschky, Bastian Widenmayer & Oliver Gassmann (2011) Frugal Innovation in Emerging Markets, Research-Technology Management, 54:4, 38-45

To link to this article: <http://dx.doi.org/10.5437/08956308X5404007>



Published online: 28 Dec 2015.



Submit your article to this journal [↗](#)



Article views: 503



View related articles [↗](#)



Citing articles: 6 View citing articles [↗](#)

FRUGAL INNOVATION IN EMERGING MARKETS

Successful frugal innovation requires a strong local presence in the emerging markets that are demanding affordable, good-enough products.

Marco Zeschky, Bastian Widenmayer, and Oliver Gassmann

OVERVIEW: *The quality and number of innovations developed by multinational companies from emerging countries is increasing dramatically. In particular, frugal innovations—“good-enough,” affordable products that meet the needs of resource-constrained consumers—have created tremendous demand in emerging markets. While the development of such products has largely been the domain of local corporations in emerging countries, Western corporations have recently started to engage in frugal innovation as well. This is a difficult task for Western firms, however, because their business models and organizational structures are traditionally designed for the development of advanced products for the affluent few at the top of the economic pyramid. Using Swiss weighing-instrument manufacturer Mettler Toledo as a case example, this article suggests that frugal innovations are largely developed by local R&D subsidiaries of Western firms in emerging countries. A substantial degree of autonomy for those local R&D subsidiaries, including product-portfolio responsibilities, can facilitate the development of frugal innovation.*

KEY CONCEPTS: *Frugal innovation, Disruptive innovation, Emerging markets, R&D organization, Subsidiaries*

Innovative ideas and products are increasingly being developed by local companies in emerging nations and marketed worldwide. Corporations from China and India in particular have flourished by developing what Christensen (1997) has characterized as disruptive innovation, products that offer superior customer value and low costs (Brown and Hagel 2005; London and Hart 2004). In 1996, for example, the Chinese home appliances firm Haier developed a washing machine called Mini Magical Child for the local market. Designed for small daily loads, it offered a real alternative to large, expensive washing machines. The Mini Magical Child was an immediate success, and a similar product based on it has now been marketed worldwide (Hang, Chen, and Subramian 2010). In 1992, the Chinese firm Galanz developed a low-cost, energy-efficient microwave small

Marco Zeschky is an assistant professor of technology management at the University of St. Gallen, Switzerland. From 1998 to 2005, he studied industrial engineering with a focus on mechanical engineering at the Darmstadt University of Technology and at the Royal Institute of Technology in Stockholm. From 2008 to 2009, he was a visiting scholar at the Wharton School in Philadelphia. He received his PhD in business administration in 2009 from the University of St. Gallen. His research focuses on international management with a special focus on international R&D management and strategic innovation management. marco.zeschky@unisg.ch

Bastian Widenmayer is a research associate and doctoral candidate at the Institute of Technology Management at the University of St. Gallen, Switzerland. His research focuses on the management of global R&D, innovation in emerging markets, and strategic innovation

management. Bastian holds a diploma in business administration from the Technische Universität München. bastian.widenmayer@unisg.ch

Oliver Gassmann is professor and managing director of the Institute of Technology Management at the University of St. Gallen, Switzerland. After completing his PhD in business administration in 1996, he lead research and advanced development efforts for Schindler Corporation in Ebikon, Switzerland. His research focuses on how companies innovate and profit from innovation and on global R&D management. His research has appeared in leading journals such as Research Policy, R&D Management, Journal of Management, Long Range Planning, International Journal of Technology Management, Journal of World Business, Management International Review, IEEE Transactions on Engineering Management, and Harvard Business Manager. oliver.gassmann@unisg.ch

DOI: 10.5437/08956308X5404007

enough to fit inside cramped Chinese kitchens. Galanz has managed to develop what had been a minor market—initially only 2 percent of households could afford a microwave—into a mass market, more than 60 percent of which it now controls, making the company one of the biggest microwave manufacturers in the world today (Ge and Ding 2008; Hart and Christensen 2002). Indian examples include Tata's Nano, a car that comes without such convenience and safety features as power steering, air conditioning, antilock braking, airbags, or a passenger-side mirror. With the Nano selling at around \$2,200, Tata is unable to meet the demand for it in India. More recently, Western corporations have also started engaging in disruptive innovation. For example, GE's healthcare division located in China developed a portable, no-frills ultrasound machine for use in rural Chinese areas.

These are all examples of “good-enough” products that meet basic needs at a low cost and thus provide high value. Products like these are at the center of Christensen's concept of low-end disruptive innovation (1997); they have also been termed “resource-constrained innovations” (Ray and Ray 2010), “cost innovations” (Williams and van Triest 2009; Williamson 2010), and “frugal innovation” (Economist 2010). We have adopted the term *frugal innovation*, defined as responding to severe resource constraints with products having extreme cost advantages compared to existing solutions. The products of frugal innovation often look inferior to existing solutions because they provide limited functionality and are often made of simpler, cheaper materials.

The examples of Haier, Galanz, Tata, and GE illustrate the typical course of frugal innovations, which are often initially developed to meet the needs of resource-constrained consumers in emerging markets. Nevertheless, these products are increasingly finding their way to markets in developed countries. The Haier washing machine and the Galanz microwave are currently being sold in the United States and Europe, and the Nano is scheduled for sale in Europe in 2012, once Tata has conformed to the minimum safety requirements stipulated by law. GE's portable ultrasound scanner has been marketed to hospitals in the United States, allowing hospitals with limited financial resources to afford diagnostic devices that meet basic requirements at a fraction of the cost of GE's standard ultrasound machines (Immelt, Govindarajan, and Trimble 2009).

In this paper, we set out to explore the question of how Western corporations are organizing frugal innovation efforts. Based on in-depth case studies from five firms, we identified patterns in the strategy and organization of successful frugal innovators. We use Mettler Toledo, a maker of weighing instruments, as an example case study illustrating the trends and patterns we found in successful frugal innovators.

The Challenges of Frugal Innovation

Due to the resource constraints typical in emerging markets, consumers in these markets are very value conscious; many have only recently shifted from being non-consumers to being consumers (Christensen 1997). An increasing segment of these markets is made up of an emerging middle class whose members live on \$2 to \$13 per day at 2005 purchasing power parity prices (Prahalad 2010). This middle class is a rapidly growing segment of the population. Projections indicate that the number of Chinese households whose income exceeds their basic needs will quadruple from around 55 million in 2008 to 212 million in 2013, with an average income of about \$5,000 per annum. In India, 5 percent of households are predicted to earn around \$4,000 per annum by 2020 (Williamson and Zeng 2009). Despite their increasing purchasing power, these new consumer households will still have little excess income and will need to take care to get value for their money. Consumers in Western markets are also becoming more value oriented, especially in the wake of the recent economic crisis. U.S. personal consumption expenditure has dropped by more than 3 percentage points during the recession (Lee, Rabanal, and Sandri 2010), and Western consumers have begun to look for simpler offerings that offer the greatest value (Flatters and Willmott 2009).

The growing market for frugal innovation brings two main challenges for Western corporations. First, for long-term success in frugal innovation, Western corporations must rethink their established business models. Typically, the business models of Western companies operating in emerging countries have not considered the resource-constrained consumer, but rather have focused on the affluent few at the top of the population pyramid who possess the buying power to afford Western products (Arnold and Quelch 1998; Prahalad and Lieberthal 1998). However, this strategy of earning high margins on a few affluent consumers is increasingly in question for two reasons. On the one hand, frugal innovations will over time also attract affluent consumers who may decide to go for less expensive products that still meet their needs. On the other hand, the growing middle class is becoming an increasingly interesting market that offers great business potential. If they continue to ignore the growing middle class in emerging markets, Western companies run the risk of losing market share to the rising competition, both in the emerging markets and in their home markets. These trends will force Western companies to adapt, fundamentally altering the way in which products and markets are defined and paired.

Second, Western companies that want to engage in frugal innovation must build organizational structures and capabilities to enable the development of frugal products. Like the business models that shape them, existing R&D processes and structures are often optimized

for the development of advanced products and technologies targeted at high-end consumers. Currently, the role of local R&D units is often still to adapt centrally developed products to local needs, thereby exploiting existing competencies and avoiding the costs of new product development (Gassmann and von Zedtwitz 1998; Pearce 1999). The resulting products often entail changes in design and the ranges of features, but they are still usually based on an advanced Western product architecture that only a few rich people can afford. Previous research, however, suggests that successful frugal innovation requires a deep understanding of the specific environment for which such products are developed. A more prominent local presence and a fundamental new-product development effort may be necessary to develop a truly effective frugal innovation process (Williamson 2010).

Research Methodology

Our research focuses on the question of how Western companies are organizing their frugal innovation activities in emerging markets. At the outset of our research, we scanned research databases and the public press to identify examples of frugal product innovations produced by firms based in developed economies. In total, we identified 13 firms representing a variety of industries. We contacted the R&D heads of these companies and asked if they were willing to provide deeper information about their organization of frugal innovation. Eight firms initially agreed to participate in our research, but we had to drop three firms due to insufficient availability of data, resulting in five usable cases: Mettler Toledo's basic weighing scale, GE's portable ultrasound machine, Logitech's M215 computer mouse, Philips's bedside patient monitoring system, and Siemens's computed tomography scanner. Besides having similar structures regarding the organization of R&D, all of these firms have a history of successful frugal innovation.

Using a qualitative case study approach (Eisenhardt 1989; Yin 2003), we collected data mainly through semistructured, in-depth interviews with top managers (CEOs, CTOs, regional or national heads) and middle-level managers (central and local R&D heads, central and local corporate technology managers). In total, we conducted 24 interviews in the five participating companies. Each interview lasted between 60 and 150 minutes. From the interview data, we developed case studies of each firm and checked with our respondents to ensure that the information presented in those case studies was correct. We kept a database of all firm information and used tabular pattern matching to identify similarities and differences in the cases (Miles and Huberman 1994). During our iterative research process, we continuously updated the database with new information.

Western companies that want to engage in frugal innovation must build organizational structures and capabilities to enable the development of frugal products.

General Patterns of Organizing Frugal Innovation

All of the firms we studied take a similar approach to organizing their frugal innovation activities (Table 1). From a strategic perspective, all firms had different triggers for engaging in frugal innovation. Pressed by increasing pressure from low-cost competition, Mettler Toledo made the strategic decision to meet its competitors head-on by launching its basic weighing scale, in spite of concerns that such a product could cannibalize the company's existing products in Western markets. Similarly, GE developed its portable ultrasound machine and marketed it in developed countries when they recognized that low-cost competitors from emerging countries would take over the resource-constrained market segment if it did not get there first. Since then, GE has adapted its product-market strategies in the United States and other developed countries to include frugal-product offerings. Philips and Siemens have chosen to make a more gradual change in their product offerings, developing frugal product lines that complement the lower end of their existing offerings and that were initially targeted only at the Chinese market. Similarly, Logitech developed the M215 computer mouse to offer a low-cost product to the Chinese market. With the exception of Mettler Toledo, all of the participating firms initially developed frugal products for China or other emerging countries and only later marketed these products in developed countries when they realized the business potential there.

From an organizational perspective, all of the frugal innovations explored in our study were developed by subsidiaries located in China. While the subsidiaries of

Table 1.—The frugal innovations included in this study were developed in the Chinese R&D subsidiaries of the Western MNCs.

	<i>Mettler Toledo Basic Weighing Scale</i>	<i>GE Portable Ultrasound Machine</i>	<i>Logitech Computer Mouse M215</i>	<i>Philips Bedside Patient Monitoring System</i>	<i>Siemens Computed Tomography Scanner</i>
Product characteristics	Limited functionality, sufficient accuracy, conventional materials, high reliability	Simple functions, standard laptop as data processing and imaging unit, support of local language, portable, easy to use	Simple functions, established technologies (e.g., USB dongle, wireless connection), basic packaging, ease of use	Simple functions, robust design, high reliability, ease of use	Simple functions, designed for intensive use and fast workflow, high reliability, ease of use
Motivation for developing product	Preempting low-cost competitors from gaining market shares	Gaining a foothold in the rural Chinese market	Offering basic low-cost product for the Chinese market	Offering a basic product to Chinese hospitals to replace local low-cost competitors	Complementing lower end of existing product lines to attract Chinese low-cost customers
Implementation of product development	Central decision, decentralized development and marketing of scale in Chinese R&D subsidiary. Initial target market: Worldwide	Decentralized decision in Chinese R&D subsidiary to develop portable ultrasound machine, central decision to take it later to developed markets. Initial target market: China	Decentralized decision to develop M215 in Chinese R&D subsidiary and to market it worldwide later. Initial target market: China	Central decision, decentralized development of monitoring system in Chinese R&D subsidiary. Central decision to take the monitoring system to developed markets. Initial target market: Emerging markets	Central decision, decentralized development of scanner in Chinese R&D subsidiary. Central decision to take the scanner to developed markets. Initial target market: Emerging markets

Mettler Toledo, GE, and Logitech enjoyed a rather high degree of autonomy from headquarters in the development of frugal products, the development activities in the Chinese subsidiaries of Philips and Siemens were more tightly controlled by central headquarters. Although local cost advantages were a persistent motivation for local development activities, the findings show that understanding the local environment and user behavior was vital to product success. Managers at Mettler Toledo, GE, Logitech, and Siemens told us that having a team consisting almost exclusively of local engineers guaranteed that they could effectively translate local requirements into final products. Logitech found through local presence that fancy product packaging suggested to Chinese consumers that the product was also expensive. Based on this insight, they reduced the packaging for the M215 to a minimum. Philips sought to understand how their products were actually used under local conditions. Through joining local doctors at work, they found out that their bedside patient monitoring system was used not only as a display device but also as a writing pad that was carried around and even taken into operating rooms where it was exposed to sweat and blood, a scenario unimaginable in Western hospitals.

The motivation and organization of frugal innovation were very similar across all of the cases that we studied.

However, Mettler Toledo was distinct in terms of its strategy and its approach to the organizational implementation of frugal innovation. Working from the strategic decision to expand its activity in the low-cost segment worldwide, the company has been engaging in frugal innovation for more than a decade. For this reason, Mettler Toledo's basic weighing scale offers an exemplary case of frugal innovation.

Mettler Toledo: Developing a Basic Scale in China

Mettler Toledo (MT) was founded in Switzerland in 1945 and is a global leader in the manufacture and marketing of precision weighing instruments for use in laboratory, industrial, and food-retailing applications. MT was the first company to develop a fully electronic balance approved for the highest accuracy class (measuring with an accuracy of up to one ten-millionth gram). With 10,000 employees worldwide, MT generated sales of about \$2 billion in 2009, of which 41 percent were in Europe, 35 percent in the Americas, and 24 percent in Asia and other regions.

Global R&D Organization

MT is a matrix organization made up of a product organization and a market organization. MT's five divisions (Laboratory, Industry, Retail, Process Analytics, and

Product Inspection), which make up the product organization, develop and manufacture precision weighing instruments. The market organization is responsible for distributing the company's products within its three main regions of operation (Europe, North America, and Asia/Pacific). Historically, MT's R&D organization has evolved partly on the basis of acquisitions made to gain new competencies and access new markets. As a result of that practice, along with the high level of autonomy accorded to MT's business units, the company's R&D organization is globally dispersed with R&D headquarters in Switzerland and R&D subsidiaries in the United States, Great Britain, Germany, and China. While the headquarters focuses on the development of core technologies for the entire organization, the business units are responsible for their own product portfolios and thus for distinct development activities in the local R&D subsidiaries.

The R&D subsidiary in China came into existence in 1987, when MT entered into a joint venture with a low-end Chinese weighing-instrument manufacturer in Shanghai to gain access to the Chinese market and to acquire know-how regarding low-cost product development. In 1999, after MT had built sufficient internal product-development and market know-how, it took over the joint venture completely and extended the Chinese unit's competencies in the development of low-cost scales.

At the beginning of 2000, MT's business in its traditional, developed-economy markets had begun to erode due to increasing competition from low-cost manufacturers, many of them from China. Realizing its rather weak position in the low-cost segment, MT decided to develop a basic, entry-level scale to improve its competitive position and regain lost market share. In 2004, based on its accumulated know-how in the area of low-cost product development, the R&D subsidiary in Shanghai, which consisted only of Chinese engineers, developed the basic scale with only minor assistance from the central R&D headquarters in Switzerland. Based on its freedom and abilities in product development, the Chinese subsidiary decided which technologies and features were included in the basic scale. In contrast to MT's existing product offerings, this basic scale was characterized by a considerably lower yet still high-quality gauging accuracy, cost-efficient materials, a different design, and the absence of advanced features such as fully automated adjusting, protective covering, dynamic weighing, electronic data analysis via PC, and programmability.

Challenges of Frugal Innovation

Rising competition from low-cost manufacturers forced MT to combine two contrary approaches for emerging markets: one that continued to profit from the established global customer base by providing advanced—and consequently more expensive—products and one that

Understanding the local environment and user behavior was vital to product success.

offered low-cost products that could have some negative effects on the advanced offerings. Considering the massive low-cost competition coming from China and the worldwide market potential of low-cost weighing scales, MT decided to take the risk of jeopardizing its existing high-end product offering in order to tap the huge potential of the low-cost market segment. As the head of R&D in MT's laboratory division told us, "We would rather cannibalize our own business than lose it altogether." Ten years later, the decision proved to be right. Today, the basic weighing scale has gained a significant market share in the low-cost product segment in China and in other emerging markets, as well as in a number of Western markets.

Naturally, the company also faced internal organizational challenges associated with the change in the product range. One arm of the company's market organization, disturbed by the significant differences between the basic, low-cost scale and MT's more-typical advanced offerings, was initially unwilling to sell the basic scale. Only when other low-cost providers in the market began gnawing at the company's profits in that region was the market organization willing to include the basic scale in its product portfolio.

Implications

The Mettler Toledo case, along with insights from GE, Logitech, Philips, and Siemens, offers important strategic and organizational lessons for Western companies pursuing frugal innovation. First, frugal innovations are homegrown products that cannot be easily derived from existing Western products. Instead, they are the result of a unique value architecture that is grounded in the drive to meet basic requirements at the lowest possible cost. Second, developing frugal innovations requires local organizational structures and resources to help develop an understanding of the needs of the resource-constrained consumers who are the target market for frugal innovations.

1. Understand the value architecture of frugal innovation.

Resource-constrained environments like those found in emerging markets present fertile ground for frugal innovations. What frugal innovations have in common is that they are all born of a situation of constraint and designed to meet the relatively basic needs of poorer consumers. To compete in this context, Western companies must change their mindset and see low-income populations as potential markets that offer great business opportunities for the right products. The development of frugal products can be supported by several design-to-cost instruments, such as value analysis, target costing, or quality function deployment. However, the key to success will be the attitude of the development team, which must be oriented toward meeting a radical cost goal.

Three factors are vital if a company is to meet the aggressive price-performance needs of resource-constrained consumers: low-cost manufacturing; simple, low-cost materials and design; and a focus on basic functionality and minimal feature sets. MT's basic scale incorporated all three of these factors. Since it was developed and manufactured entirely in China, the scale achieved cost advantages in manufacturing. The scale incorporates more cost-efficient materials, relying on high-quality synthetic materials in place of the metal construction common in MT's higher-end product lines. Finally, the basic scale features only the most basic functions and design elements and offers high-quality but good-enough gauging accuracy that meets local use requirements.

To achieve these requirements, MT relied on the low-cost competence of its Chinese engineers, who were able to assemble a high-quality but good-enough product that met the basic measuring requirements of its target market. The basic scale's cost advantage is a result of reduced development and manufacturing costs and savings due to the use of different materials and the inclusion of fewer features. Other frugal products build on the same cost structure. For example, GE's ultrasound machine is largely based on commodity technologies (for example, notebook screens for imaging and common interfaces between computer hardware and peripheral equipment) and was developed and manufactured in GE's Chinese subsidiary in Shanghai.

2. Establish local organizational structures to enable frugal innovation.

Most Western companies maintain local R&D units in emerging countries in order to profit from direct access to the market and access to local talent. Local R&D units also enable them to build a presence in the hot spots where new technological developments are originating (Gassmann and Han 2004). However, the value

**Frugal innovations
are the result of a
unique value
architecture that is
grounded in the
drive to meet basic
requirements at the
lowest possible cost.**

architecture of frugal innovations requires development capabilities that are difficult, if not impossible, to build in central headquarters. To build such capabilities effectively, Western companies should empower local R&D subsidiaries to access the power of local knowledge.

Autonomy from central R&D headquarters seems to be a necessary but not sufficient condition to enable the local development of frugal innovations. The subsidiary must also be able to understand local needs and translate those into appropriate product solutions. MT's entry-level scale was to a large degree developed by the Chinese R&D subsidiary, comprising solely Chinese engineers. Having local people on the team facilitated access to local customers and ensured an understanding of the essential functions those customers required. Similarly, in emerging markets, GE employs local growth teams made up of local workers who know the market and are able to communicate market needs to the product-development team. The Chinese R&D teams of Siemens and Philips also consist almost exclusively of local engineers who collaborate with local doctors and institutions. Based on this collaboration, these teams have developed products that fit into tiny rooms, are robust enough to handle dirt and frequent power fluctuations, and can endure intense usage.

Building effective capabilities for frugal innovation depends largely on the ability of the R&D team to sense local needs and translate them into effective, low-cost products. Since the living conditions, culture, value system, and overall societal relationships in emerging markets are very different from those in developed countries, it is often difficult, if not impossible, for engineers raised

and trained in the West to fully understand the needs of foreign customers. Prior research has shown the importance of a firm's absorptive capacity to sense and use information to improve its R&D competencies (Cohen and Levinthal 1990). Successful frugal innovation is facilitated by having on the R&D team local people who bring personal experience with the environment in which the product will be used.

Conclusion

Competition from low-cost producers, many of them in emerging markets, is entering a new phase as companies from emerging markets have begun to export to developed markets homegrown products designed to meet local needs. But Western companies should embrace frugal innovation as an opportunity rather than fear it as a threat. Although frugal innovations might compete against existing products in both emerging and developed markets, the potential for massive low-end profits may well outweigh losses in higher-end product lines. Western companies should therefore consider adapting existing innovation strategies to the development of frugal innovation.

Successful frugal innovation begins in the mind. Before Western MNCs can effectively engage in frugal innovation, they must understand the unique characteristics of frugal products: very high customer benefits at very low costs. The path to successful frugal innovation also demands attention to promoting a mindset of frugality within R&D teams and establishing the necessary structures by empowering R&D subsidiaries in emerging markets.

The way Western companies engage in frugal innovation has wider implications for globally dispersed R&D and business in general. The case insights presented in this paper imply that power and responsibilities in the future will be increasingly shifted to the periphery where unique frugal-innovation competencies are grown. Central R&D headquarters are likely to take on the role of network coordinators, bundling individual competencies in the network to initiate new development activities anywhere in the world. In this regard, the ability to develop frugal innovation represents a specific and significant competency. Still, the Western MNC as a parent company will play an important role; the subsidiaries developing frugal innovation will profit from the firm's global brand and worldwide sales channels, which may provide a competitive advantage over emerging low-cost competitors.

References

Arnold, D. J., and Quelch, J. A. 1998. New strategies in emerging markets. *Sloan Management Review* 40(1): 7–20.
Brown, J. S., and Hagel, J. 2005. Innovation blowback: Disruptive management practices from Asia. *McKinsey Quarterly* 1:35–45.
Christensen, C. M. 1997. *The Innovator's Dilemma: When New Technologies Cause Great Firms to Fail*. Boston, MA: Harvard Business School Press.

Building effective capabilities for frugal innovation depends largely on the ability of the R&D team to sense local needs and translate them into effective, low-cost products.

Cohen, W. M., and Levinthal, D. A. 1990. Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly* 35(1): 128–152.
Economist. 2010. First break all the rules: The charms of frugal innovation. Special report on innovation in emerging markets. *The Economist*, April 17: 3–5. <http://www.economist.com/node/15879359> (accessed April 7, 2011).
Eisenhardt, K. M. 1989. Building theories from case study research. *Academy of Management Review* 14(4): 532–550.
Flatters, P., and Willmott, M. 2009. Understanding the post-recession consumer. *Harvard Business Review* 87(7/8): 106–112.
Gassmann, O., and Han, Z. 2004. Motivations and barriers of foreign R&D activities in China. *R & D Management* 34(4): 423–437.
Gassmann, O., and von Zedtwitz, M. 1998. Organization of industrial R&D on a global scale. *R & D Management* 28(3): 147–161.
Ge, G., and Ding, D. 2008. A strategic analysis of surging Chinese manufacturers: The case of Galanz. *Asia Pacific Journal of Management* 25(4): 667–683.
Hang, C.-C., Chen, J., and Subramian, A. M. 2010. Developing disruptive products for emerging economies: Lessons from Asian cases. *Research-Technology Management* 53(4): 21–26.
Hart, S., and Christensen, C. M. 2002. The great leap: Driving innovation from the base of the pyramid. *MIT Sloan Management Review* 44(1): 51–56.
Immelt, J. R., Govindarajan, V., and Trimble, C. 2009. How GE is disrupting itself. *Harvard Business Review* 87(10): 56–65.
Lee, J., Rabanal P., and Sandri D. 2010. *US Consumption after the 2008 Crisis*. Staff Position Note No. 2010/01. International Monetary Fund, January 15.
London, T., and Hart, S. L. 2004. Reinventing strategies for emerging markets: Beyond the transnational model. *Journal of International Business Studies* 35(5): 350–370.
Miles, M. B., and Huberman, A. M. 1994. *Qualitative Data Analysis: An Expanded Sourcebook*. 2nd ed. Thousand Oaks, CA: Sage Publications.
Pearce, R. D. 1999. Decentralised R&D and strategic competitiveness: Globalised approaches to generation and use of technology in multinational enterprises (MNEs). *Research Policy* 28(2–3): 157–178.

- Prahalad, C. K. 2010. *The Fortune at the Bottom of the Pyramid: Eradicating Poverty Through Profits*. 5th ed. Upper Saddle River, NJ: Wharton School Publishing.
- Prahalad, C. K., and Lieberthal, K. 1998. The end of corporate imperialism. *Harvard Business Review* 76(4): 68–79.
- Ray, P. K., and Ray, S. 2010. Resource-constrained innovation for emerging economies: The case of the Indian telecommunications industry. *IEEE Transactions on Engineering Management* 57(1): 144–156.
- Williams, C., and van Triest, S. 2009. The impact of corporate and national cultures on decentralization in multinational corporations. *International Business Review* 18(2): 156–167.
- Williamson, P. J. 2010. Cost innovation: Preparing for a “value-for-money” revolution. *Long Range Planning* 43(2–3): 343–353.
- Williamson, P. J., and Zeng, M. 2009. Value-for-money strategies for recessionary times. *Harvard Business Review* 87(3): 66–74.
- Yin, R. K. 2003. *Case Study Research: Design and Methods*. 3rd ed. Thousand Oaks, CA: Sage Publications.

Reprints

THE ART OF TECHNOLOGY MANAGEMENT

Forty-nine selections the “Managers at Work” department of RESEARCH-TECHNOLOGY MANAGEMENT are now available in paperback. To order, see inside back cover.

Leadership and R&D Productivity
 To Motivate, Set Goals
 Motivating Research Scientists to Reach for the Eagles
 Wining the Respect of Subordinates
 How Am I Doing?
 Become A Better Coach
 Exercising Leadership in the Corporation
 What Does the CEO Want?
 Are You Credible With Your CEO?
 Encouraging ‘Little C’ and ‘Big C’ Creativity
 Fostering Breakthrough Innovations
 What to do About ‘NIH’
 To Innovate Faster, Try the Skunk Works
 Overcoming Roadblocks to
 Commercializing Industrial R&D Projects
 Overcoming Roadblocks to the Marketplace
 Technology Transfer: A GM Manager’s Strategy

Metanoic Society Helps Shell Commercialize Product Ideas in Half the Time
 Making Customer Visits Pay
 Teams Speed Commercialization of R&D Projects
 Building Teams—What Works (Sometimes) Before you Try Team Building
 Creating High-Performance Teams
 Rules of Thumb for Project Management
 Four Steps to Better Staff Meetings
 Red Flags at Dawn, or Predicting R&D Project Termination at Start-Up
 Hiring People Who Do Good Research
 Revisiting the Dual Ladder at General Mills
 Misusing the Dual Ladder, or the ‘The Case of Joe Mertz’
 Building Careers at Procter & Gamble
 ...AND MORE